

How Vision Transformer (ViT) Works

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1 Introduction

Vision Transformer (ViT) fundamentally adapts the Transformer architecture from Natural Language Processing (NLP) to computer vision. The core idea is to divide an image into small patches and process them similarly to word tokens in a sentence. Unlike traditional Convolutional Neural Networks (CNNs), which progressively extract local features using convolution kernels, ViT directly models global interactions through self-attention.

2 Overall Pipeline

Image

→ Patch Partition

→ Patch Embedding

→ Position Embedding

→ Transformer Encoder

→ Classification Head

3 Patch Embedding

Given an image of size: $224 \times 224 \times 3$ and the image is divided into: $14 \times 14 = 196$ patches. Each patch has shape: $16 \times 16 \times 3$. After flattening: $16 \times 16 \times 3 = 768$ Thus, each patch becomes a 768-dimensional vector.

However, the flattened vector is still only raw RGB pixel information:

$$[pixel_1^R, pixel_1^G, pixel_1^B, pixel_2^R, \dots]$$

It does not yet contain semantic structures such as edges, textures, or shapes.

Therefore, ViT applies a learnable linear projection:

$$y = Wx + b$$

where:

$$W \in R^{768 \times 768}$$

Although the dimensionality remains unchanged, the representation space changes substantially. The projection maps the raw pixel space into a semantic embedding space that is more suitable for attention-based learning.

The projection layer learns feature combinations such as:

- edges
- textures
- brightness contrasts
- local visual patterns

This role is conceptually similar to the first convolution layer in CNNs.

Interestingly, the patch embedding operation can also be implemented as:

```
Conv2D(  
    kernel_size = patch_size,  
    stride = patch_size,  
    out_channels = embedding_dim  
)
```

which is commonly used in practical ViT implementations.

4 Why Position Embedding is Necessary

Transformer architectures do not inherently contain spatial or sequential inductive biases. CNNs naturally preserve spatial locality because convolution kernels slide across neighboring pixels. In contrast, self-attention alone has no explicit notion of spatial position.

Therefore, ViT introduces position embeddings to encode spatial information.

The actual Transformer input is:

$$h_i = z_i + p_i$$

where:

- z_i is the patch embedding
- p_i is the position embedding

The addition is pair-wise and element-wise.

Position embeddings do not overwrite semantic content. Instead, they introduce spatial biases into the embedding geometry. In practice:

$$\|z_i\| \gg \|p_i\|$$

Thus, content information generally dominates while positional information acts as a directional perturbation.

This allows the Transformer to preserve **semantic similarity** and **spatial distinction** simultaneously. For example, two visually similar lung patches located in different regions of a chest X-ray can still maintain high semantic similarity while remaining spatially distinguishable.

5 Self-Attention

The core mechanism of ViT is self-attention. Unlike CNNs, where receptive fields gradually expand layer by layer, self-attention enables direct global interaction between arbitrary patches from the very first layer.

The self-attention operation is:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

This allows each patch token to attend to all other patches.

Consequently, ViT naturally models:

- long-range dependencies
- global context
- cross-region relationships

which are especially important in medical imaging tasks.

6 How Q, K, and V are Generated

For each token embedding:

$$h_i \in R^{768}$$

the Transformer applies three learnable linear projections:

$$Q_i = h_i W_Q$$

$$K_i = h_i W_K$$

$$V_i = h_i W_V$$

where:

$$W_Q, W_K, W_V$$

are learnable matrices.

These projections define different latent relation spaces rather than merely computing raw similarity.

Conceptually:

- Query (Q): what the token is searching for
- Key (K): what the token can provide
- Value (V): the information being transmitted

The attention score between token i and token j is:

$$a_{ij} = Q_i K_j^T$$

After softmax normalization:

$$\alpha_{ij} = \text{Softmax}(a_{ij})$$

the final output is:

$$o_i = \sum_j \alpha_{ij} V_j$$

Thus, each token becomes context-aware by aggregating information from other tokens.

7 Multi-Head Attention

ViT typically uses multi-head attention.

For example:

$$768 = 12 \times 64$$

This does not mean that the original 768 dimensions are simply partitioned into disjoint blocks. Instead, the entire embedding first undergoes learnable linear projection:

$$Q = XW_Q$$

before being reshaped into multiple heads.

Therefore, each attention head observes the full input embedding through a different learned projection space.

Different heads can specialize in different relation patterns, such as:

- spatial structure
- symmetry
- texture
- semantic alignment

This is better understood as:

representation diversification

rather than simple feature partitioning.

After attention computation, all heads are concatenated and projected again:

$$\text{MultiHead}(X) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W_O$$

where:

$$W_O$$

acts as a communication layer that mixes information across heads.

8 CLS Token in Multimodal Fusion

ViT introduces a special:

$$[CLS]$$

token for image classification.

The final CLS embedding represents the global image representation.

However, in multimodal Transformer systems, token-level interactions are often more important than a single global vector. For example, in chest X-ray and clinical text fusion:

`text token "effusion" → attends to specific lung-region patches`

Therefore, many multimodal architectures preserve patch-level tokens instead of relying solely on the CLS token.

9 Cross Attention in Multimodal Learning

Cross attention extends self-attention by using:

- Query from one modality
- Key and Value from another modality

For example:

$$Q_{text} = text \cdot W_Q$$

$$K_{image} = image \cdot W_K$$

$$V_{image} = image \cdot W_V$$

This allows text tokens to query image patches directly.

Consequently, multimodal Transformers naturally support:

- image-text alignment
- cross-modal interaction
- interpretable attention maps

10 Why ViT is Important in Medical Imaging

Medical images often require:

- global structural reasoning
- long-range dependency modeling
- multi-region interaction

Examples include:

- cardiomegaly
- pleural effusion
- bilateral lung patterns
- distant lesion correlation

Since self-attention directly models global dependencies, ViT has become increasingly important in:

- radiology
- pathology
- surgical video analysis
- OCT imaging
- fundus imaging

11 ViT and Multimodal Vision-Language Models

Image patches and text tokens can both be represented as Transformer token sequences.

Therefore, many multimodal models naturally build upon ViT-based architectures, including:

- CLIP
- BLIP
- MedCLIP
- multimodal Transformers

In our chest X-ray + ClinicalBERT project, the image pipeline is:

```
X-ray image
→ patch embeddings
→ Transformer encoder
→ image feature sequence
```

Clinical text embeddings are then fused with image tokens through cross attention, forming a simplified Vision-Language Transformer framework.